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**2** A particle $P$ of mass $m$ is attached to one end of a light elastic string of natural length $a$ and modulus of elasticity $2mg$. A particle $Q$ of mass $km$ is attached to the other end of the string. Particle $P$ lies on a smooth horizontal table. The string has part of its length in contact with the table and then passes through a small smooth hole $H$ in the table.

Particle $P$ moves in a horizontal circle on the surface of the table with constant speed $\sqrt{\frac{1}{2}ga}$. Particle $Q$ hangs in equilibrium vertically below the hole with $HQ = \frac{1}{4}a$.

**(a)** Find, in terms of $a$, the extension in the string. [4]

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**(b)** Find the value of $k$. [2]

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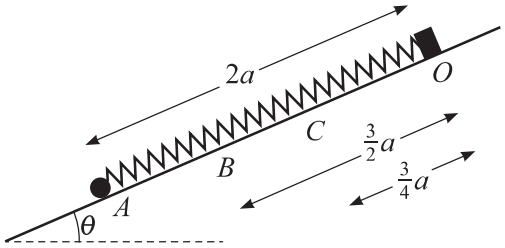


A light spring of natural length $a$ and modulus of elasticity $kmg$ is attached to a fixed point $O$ on a smooth plane inclined to the horizontal at an angle $\theta$, where $\sin\theta = \frac{3}{4}$. A particle of mass $m$ is attached to the lower end of the spring and is held at the point $A$ on the plane, where $OA = 2a$ and $OA$ is along a line of greatest slope of the plane (see diagram).

The particle is released from rest and is moving with speed $V$ when it passes through the point $B$ on the plane, where $OB = \frac{3}{2}a$. The speed of the particle is $\frac{1}{2}V$ when it passes through the point $C$ on the plane, where $OC = \frac{3}{4}a$.

Find the value of $k$. [7]

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